

BABU BANARASI DAS

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SCHOOL OF COMPUTER APPLICATIONS

NoSQL Project On

Basketball Match Management System Using MongoDB

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ACKNOWLEDGEMENT

I sincerely express my gratitude to my teacher Mr. Harendra Singh for giving me the opportunity to work on this NoSQL project titled “Basketball Match Management System Using MongoDB”.

This project helped me understand the basic concepts of NoSQL databases and learn how MongoDB is used to create, retrieve, update, sort and delete data. I also learned how different query operators such as greater than, less than, equal to, AND, and OR are used in MongoDB.

I am thankful to my teacher for the guidance and support provided throughout the completion of this project.

INTRODUCTION

MongoDB is a popular NoSQL database management system that stores data in the form of documents. Unlike traditional relational databases, MongoDB does not require data to be stored in fixed rows and columns.

In this project, a Basketball Match Management System has been created using MongoDB. The database contains information about 50 basketball matches, including their match ID, teams, scores, and venue.

Different MongoDB operations have been performed on the basketball match data, such as inserting, displaying, updating, deleting, sorting, counting, and searching records. Comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte have also been demonstrated.

OBJECTIVES

- To understand the concept of NoSQL databases.
- To learn the basic features of MongoDB.
- To create a database and collection.
- To insert 50 basketball match records.
- To perform CRUD operations.
- To use comparison operators.
- To perform sorting and filtering.
- To understand AND and OR operations.

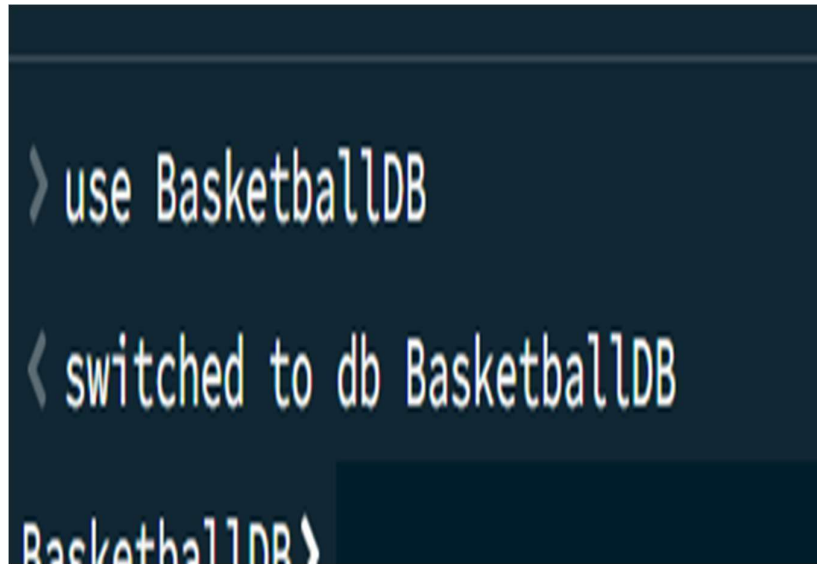
1.

CREATE DATABASE

MongoDB Query: use

BasketballDB

Explanation: Creates or switches to the BasketballDB database.



```
> use BasketballDB
< switched to db BasketballDB
BasketballDB>
```

CREATE COLLECTION

MongoDB Query:

db.createCollection("matches")

Explanation: Creates a collection named matches.

2.

```
> db.createCollection("matches")  
< { ok: 1 }  
BasketballDB>
```

INSERT 50 BASKETBALL MATCH RECORDS

MongoDB Query:

```
db.matches.insertMany([  
  {matchId:1, team1:"India", team2:"Australia", team1Score:92, team2Score:87, venue:"New  
Delhi"},  
  {matchId:2, team1:"USA", team2:"Canada", team1Score:105, team2Score:98, venue:"Toronto"},  
  {matchId:3, team1:"Spain", team2:"France", team1Score:88, team2Score:91, venue:"Madrid"},  
  {matchId:4, team1:"Germany", team2:"Italy", team1Score:96, team2Score:82, venue:"Berlin"},  
  {matchId:5, team1:"India", team2:"Japan", team1Score:101, team2Score:95, venue:"Mumbai"},  
  {matchId:6, team1:"Brazil", team2:"Argentina", team1Score:89, team2Score:94, venue:"Rio de  
Janeiro"},  
  {matchId:7, team1:"Australia", team2:"China", team1Score:97, team2Score:84, venue:"Sydney"},  
  {matchId:8, team1:"France", team2:"Belgium", team1Score:91, team2Score:79, venue:"Paris"},  
  {matchId:9, team1:"USA", team2:"Mexico", team1Score:112, team2Score:89, venue:"Dallas"},  
])
```

3.

```
{matchId:10, team1:"Canada", team2:"Brazil", team1Score:86, team2Score:90,  
venue:"Vancouver"},  
  
{matchId:11, team1:"India", team2:"China", team1Score:98, team2Score:93, venue:"Kolkata"},  
{matchId:12, team1:"Japan", team2:"South Korea", team1Score:87, team2Score:82,  
venue:"Tokyo"},  
{matchId:13, team1:"Spain", team2:"Germany", team1Score:94, team2Score:89,  
venue:"Barcelona"},  
{matchId:14, team1:"Italy", team2:"Greece", team1Score:81, team2Score:88, venue:"Rome"},  
{matchId:15, team1:"Australia", team2:"Japan", team1Score:103, team2Score:91,  
venue:"Melbourne"},  
{matchId:16, team1:"Argentina", team2:"Chile", team1Score:95, team2Score:78, venue:"Buenos  
Aires"},  
{matchId:17, team1:"USA", team2:"France", team1Score:108, team2Score:102, venue:"New  
York"},  
{matchId:18, team1:"Canada", team2:"Spain", team1Score:90, team2Score:97,  
venue:"Montreal"},  
{matchId:19, team1:"India", team2:"South Korea", team1Score:85, team2Score:80,  
venue:"Bengaluru"},  
{matchId:20, team1:"Brazil", team2:"Uruguay", team1Score:92, team2Score:76, venue:"Sao  
Paulo"},
```

{matchId:21, team1:"Germany", team2:"France", team1Score:99, team2Score:101, venue:"Munich"},
{matchId:22, team1:"Australia", team2:"India", team1Score:94, team2Score:89, venue:"Sydney"},
{matchId:23, team1:"USA", team2:"Spain", team1Score:115, team2Score:108, venue:"Los Angeles"},
{matchId:24, team1:"Japan", team2:"China", team1Score:83, team2Score:96, venue:"Osaka"},
{matchId:25, team1:"Argentina", team2:"Brazil", team1Score:91, team2Score:87, venue:"Cordoba"},
{matchId:26, team1:"France", team2:"Italy", team1Score:104, team2Score:92, venue:"Lyon"},
{matchId:27, team1:"Canada", team2:"USA", team1Score:88, team2Score:109, venue:"Toronto"},
{matchId:28, team1:"India", team2:"Nepal", team1Score:110, team2Score:72, venue:"New Delhi"},
{matchId:29, team1:"Spain", team2:"Portugal", team1Score:93, team2Score:85, venue:"Madrid"},
{matchId:30, team1:"Germany", team2:"Netherlands", team1Score:87, team2Score:90, venue:"Hamburg"},

{matchId:31, team1:"Australia", team2:"New Zealand", team1Score:102, team2Score:96, venue:"Perth"},
{matchId:32, team1:"India", team2:"Sri Lanka", team1Score:99, team2Score:76, venue:"Chennai"},
{matchId:33, team1:"USA", team2:"Argentina", team1Score:118, team2Score:97, venue:"Chicago"},
{matchId:34, team1:"France", team2:"Spain", team1Score:86, team2Score:93, venue:"Paris"},
{matchId:35, team1:"Brazil", team2:"Colombia", team1Score:100, team2Score:84, venue:"Brasilia"},
{matchId:36, team1:"Japan", team2:"Philippines", team1Score:89, team2Score:94, venue:"Tokyo"},
{matchId:37, team1:"Germany", team2:"Belgium", team1Score:105, team2Score:88, venue:"Frankfurt"},
{matchId:38, team1:"Canada", team2:"Mexico", team1Score:96, team2Score:81, venue:"Vancouver"},
{matchId:39, team1:"India", team2:"Iran", team1Score:91, team2Score:86, venue:"Hyderabad"},
{matchId:40, team1:"Australia", team2:"South Korea", team1Score:108, team2Score:92, venue:"Melbourne"},

{matchId:41, team1:"USA", team2:"Germany", team1Score:111, team2Score:95, venue:"Boston"},
{matchId:42, team1:"Spain", team2:"Greece", team1Score:97, team2Score:89, venue:"Valencia"},
{matchId:43, team1:"India", team2:"Japan", team1Score:102, team2Score:88, venue:"Pune"},

```
{matchId:44, team1:"Brazil", team2:"Chile", team1Score:93, team2Score:79, venue:"Rio de
Janeiro"},
{matchId:45, team1:"France", team2:"Germany", team1Score:100, team2Score:98,
venue:"Marseille"},
{matchId:46, team1:"Australia", team2:"Philippines", team1Score:107, team2Score:99,
venue:"Brisbane"},
{matchId:47, team1:"Canada", team2:"Argentina", team1Score:89, team2Score:92,
venue:"Ottawa"},
{matchId:48, team1:"India", team2:"Australia", team1Score:106, team2Score:101,
venue:"Ahmedabad"},
{matchId:49, team1:"USA", team2:"Canada", team1Score:120, team2Score:104, venue:"Las
Vegas"},
{matchId:50, team1:"Spain", team2:"Italy", team1Score:98, team2Score:83, venue:"Seville"}
]
```

Explanation: Inserts 50 basketball match documents into the matches collection

MONGOSH

```
'27': ObjectId('6ac69002a44099e363056789'),
'28': ObjectId('6ac69002a44099e36305678a'),
'29': ObjectId('6ac69002a44099e36305678b'),
'30': ObjectId('6ac69002a44099e36305678c'),
'31': ObjectId('6ac69002a44099e36305678d'),
'32': ObjectId('6ac69002a44099e36305678e'),
'33': ObjectId('6ac69002a44099e36305678f'),
'34': ObjectId('6ac69002a44099e363056790'),
'35': ObjectId('6ac69002a44099e363056791'),
'36': ObjectId('6ac69002a44099e363056792'),
'37': ObjectId('6ac69002a44099e363056793'),
'38': ObjectId('6ac69002a44099e363056794'),
'39': ObjectId('6ac69002a44099e363056795'),
'40': ObjectId('6ac69002a44099e363056796'),
'41': ObjectId('6ac69002a44099e363056797'),
'42': ObjectId('6ac69002a44099e363056798'),
'43': ObjectId('6ac69002a44099e363056799'),
'44': ObjectId('6ac69002a44099e36305679a'),
'45': ObjectId('6ac69002a44099e36305679b'),
'46': ObjectId('6ac69002a44099e36305679c'),
'47': ObjectId('6ac69002a44099e36305679d'),
'48': ObjectId('6ac69002a44099e36305679e'),
'49': ObjectId('6ac69002a44099e36305679f')
```

4. DISPLAY ALL PLAYERS

MongoDB Query:

```
db.matches.find()
```

Explanation: Retrieves all documents from the matches collection.

```

> db.matches.find()
< {
  _id: ObjectId('6ac69002a44099e36305676e'),
  matchId: 1,
  team1: 'India',
  team2: 'Australia',
  team1Score: 92,
  team2Score: 87,
  venue: 'New Delhi'
}
{
  _id: ObjectId('6ac69002a44099e36305676f'),
  matchId: 2,
  team1: 'USA',
  team2: 'Canada',
  team1Score: 105,
  team2Score: 98,
  venue: 'Toronto'
}

```

5. EQUAL TO — \$eq

MongoDB Query:

```

db.matches.find({ team1Score:
{$eq: 92}
})

```

Explanation: Finds basketball matches where Team 1 scored exactly 92 points.

```
type --> for more
> db.matches.find({
  team1Score: {$eq: 92}
})
< {
  _id: ObjectId('6ac69002a44099e36...
  matchId: 1,
  team1: 'India',
  team2: 'Australia',
  team1Score: 92,
  team2Score: 87,
  venue: 'New Delhi'
}
{
  _id: ObjectId('6ac69002a44099e36...
  matchId: 20,
  team1: 'Brazil',
  team2: 'Uruguay',
  team1Score: 92,
```

6. GREATER THAN — \$gt

MongoDB Query:

```
db.matches.find({ team1Score:
{$gt: 100}
})
```

Explanation: Finds basketball matches where Team 1 scored more than 100 points.

```
> db.matches.find({
  team1Score: {$gt: 100}
})
{
  _id: ObjectId('6ac69002a44099e36305676f'),
  matchId: 2,
  team1: 'USA',
  team2: 'Canada',
  team1Score: 105,
  team2Score: 98,
  venue: 'Toronto'
}
{
  _id: ObjectId('6ac69002a44099e363056772'),
  matchId: 5,
  team1: 'India',
  team2: 'Japan',
  team1Score: 101,
  team2Score: 95,
  venue: 'Mumbai'
}
```

7. LESS THAN — \$lt

MongoDB Query: db.matches.find({
 team1Score: {\$lt: 90}
})

Explanation: Finds basketball matches where Team 1 scored less than 90 points.

```
> db.matches.find({
  team1Score: {$lt: 90}
})
< {
  _id: ObjectId('6ac69002a44099e363056770'),
  matchId: 3,
  team1: 'Spain',
  team2: 'France',
  team1Score: 88,
  team2Score: 91,
  venue: 'Madrid'
}
{
  _id: ObjectId('6ac69002a44099e363056773'),
  matchId: 6,
  team1: 'Brazil',
  team2: 'Argentina',
  team1Score: 89,
  team2Score: 94,
  venue: 'Rio de Janeiro'
}
{
```


8.

GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

```
db.matches.find({ team1Score: {$gte: 100} })
```

Explanation: Finds basketball matches where Team 1 scored 100 or more points.

```
venue: 'Ottawa'
}
> db.matches.find({
  team1Score: {$gte: 100}
})
< {
  _id: ObjectId('6ac69002a44099e36305676f'),
  matchId: 2,
  team1: 'USA',
  team2: 'Canada',
  team1Score: 105,
  team2Score: 98,
  venue: 'Toronto'
}
{
  _id: ObjectId('6ac69002a44099e363056772'),
  matchId: 5,
  team1: 'India',
  team2: 'Japan',
  team1Score: 101,
  team2Score: 95,
  venue: 'Mumbai'
}
{
```

9.

LESS THAN OR EQUAL TO — \$lte

MongoDB Query: `db.matches.find({
 team1Score: {$lte: 90}
})`

Explanation: Finds basketball matches where Team 1 scored 90 or fewer points.

```
}  
> db.matches.find({  
  team1Score: {$lte: 90}  
})  
< {  
  _id: ObjectId('6ac69002a44099e363056770'),  
  matchId: 3,  
  team1: 'Spain',  
  team2: 'France',  
  team1Score: 88,  
  team2Score: 91,  
  venue: 'Madrid'  
}  
{  
  _id: ObjectId('6ac69002a44099e363056773'),  
  matchId: 6,  
  team1: 'Brazil',  
  team2: 'Argentina',  
  team1Score: 89,  
  team2Score: 94,  
  venue: 'Rio de Janeiro'  
}  
{
```

UPDATE OPERATION

10.

MongoDB Query: `db.matches.updateOne(
 {matchId:5},
 {$set:{team1Score:105}}
)`

Explanation: Updates the Team 1 score to 105 for the match with matchId 5.

```
}  
> db.matches.updateOne(  
  {matchId:5},  
  {$set:{team1Score:105}}  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

DELETE ONE OPERATION

MongoDB Query:

```
db.matches.deleteOne({ matchId:10  
})
```

11.

Explanation: Deletes one document matching matchId 10.

```
}  
> db.matches.deleteOne({  
  matchId:10  
})  
< {  
  acknowledged: true,  
  deletedCount: 1  
}  
BasketballDB>
```

DELETE MANY OPERATION

MongoDB Query:

```
db.matches.deleteMany({  
  
  team1Score: {$lt:90}  
  
})
```

Explanation: Deletes all basketball matches where Team 1 scored below 90 points.

12.

```
    }  
    > db.matches.deleteMany({  
      team1Score: {$lt:90}  
    })  
    < {  
      acknowledged: true,  
      deletedCount: 11  
    }  
BasketballDB>
```

13. AND OPERATION — \$and

MongoDB Query:

```
db.matches.find({  
  
  $and:[  
  
    {team1Score:{$gt:100}},  
  
    {team1:"India"}  
  
  ]  
  
})
```

Explanation: Finds matches satisfying both conditions: Team 1 is India and Team 1 scored more than 100 points.

```
}  
db.matches.find({  
  $and:[  
    {team1Score:{$gt:100}},  
    {team1:"India"}  
  ]  
})  
{  
  _id: ObjectId('6ac69002a44099e363056772'),  
  matchId: 5,  
  team1: 'India',  
  team2: 'Japan',  
  team1Score: 105,  
  team2Score: 95,  
  venue: 'Mumbai'  
}  
{  
  _id: ObjectId('6ac69002a44099e363056789'),  
  matchId: 28,  
  team1: 'India',  
  team2: 'Nepal',  
  team1Score: 110,  
  team2Score: 72,  
  venue: 'New Delhi'  
}  
f
```

14. OR OPERATION — \$or

MongoDB Query:

```
db.matches.find({  
  $or:[  
    {team1:"India"},  
    {team1:"USA"}  
  ]  
})
```

Explanation: Finds basketball matches where either India or USA is Team 1.

```
> db.matches.find({  
  $or:[  
    {team1:"India"},  
    {team1:"USA"}  
  ]  
})  
{  
  _id: ObjectId('6ac69002a44099e36305676e'),  
  matchId: 1,  
  team1: 'India',  
  team2: 'Australia',  
  team1Score: 92,  
  team2Score: 87,  
  venue: 'New Delhi'  
}  
{  
  _id: ObjectId('6ac69002a44099e36305676f'),  
  matchId: 2,  
  team1: 'USA',  
  team2: 'Canada',  
  team1Score: 105,  
  team2Score: 98,  
  venue: 'Toronto'  
}
```

15. SORT OPERATION

A. Ascending Order MongoDB Query:

```
db.matches.find().sort({  
  team1Score: 1  
})
```

Explanation: Sorts matches by Team 1's score from lowest to highest.
Use 1 for ascending order.

```
}  
> db.matches.find().sort({  
  team1Score: 1  
})  
< {  
  _id: ObjectId('6ac69002a44099e36305677f'),  
  matchId: 18,  
  team1: 'Canada',  
  team2: 'Spain',  
  team1Score: 90,  
  team2Score: 97,  
  venue: 'Montreal'  
}  
{  
  _id: ObjectId('6ac69002a44099e363056775'),  
  matchId: 8,  
  team1: 'France',  
  team2: 'Belgium',  
  team1Score: 91,  
  team2Score: 79,  
  venue: 'Paris'  
}  
{
```

B. Descending Order

MongoDB Query:

```
db.matches.find().sort({  
  team1Score: -1  
})
```

Explanation: Sorts matches by Team 1's score from highest to lowest. Use -1 for descending order.

```
Type "it" for more
> db.matches.find().sort({
  team1Score: -1
})
< {
  _id: ObjectId('6ac69002a44099e36305679e'),
  matchId: 49,
  team1: 'USA',
  team2: 'Canada',
  team1Score: 120,
  team2Score: 104,
  venue: 'Las Vegas'
}
{
  _id: ObjectId('6ac69002a44099e36305678e'),
  matchId: 33,
  team1: 'USA',
  team2: 'Argentina',
  team1Score: 118,
  team2Score: 97,
  venue: 'Chicago'
}
{
  _id: ObjectId('6ac69002a44099e363056784'),
```

17. SEARCH BY TEAM MongoDB Query:

`db.matches.find({ team1:"India" })`

Explanation: Finds all basketball matches where India is Team 1.

```
Type "it" for more
> db.matches.find({
  team1:"India"
})
< {
  _id: ObjectId('6ac69002a44099e36305676e'),
  matchId: 1,
  team1: 'India',
  team2: 'Australia',
  team1Score: 92,
  team2Score: 87,
  venue: 'New Delhi'
}
{
  _id: ObjectId('6ac69002a44099e363056772'),
  matchId: 5,
  team1: 'India',
  team2: 'Japan',
  team1Score: 105,
  team2Score: 95,
  venue: 'Mumbai'
}
```

18. COUNT DOCUMENTS MongoDB

Query:

`db.matches.countDocuments()`

Explanation: Counts the documents currently present in the collection after the delete operations.

```
> db.matches.countDocuments()
```

```
< 38
```

```
BasketballDB>
```

MONGODB OPERATORS USED

Operator	Meaning	Example
\$eq	Equal to	team1Score: {\$eq: 92}
\$gt	Greater than	team1Score: {\$gt: 100}
\$lt	Less than	team1Score: {\$lt: 90}
\$gte	Greater than or equal to	team1Score: {\$gte: 100}
\$lte	Less than or equal to	team1Score: {\$lte: 90}
\$and	All conditions true	\$and:[{team1Score:{\$gt:100}} , {team1:"India"}]
\$or	At least one condition true	\$or:[{team1:"India"}, {team1:"USA"}]

CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. A Basketball Match Management System containing information about 50 basketball matches was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update, and delete documents. I also learned how to arrange basketball match data in ascending and descending order, how comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte work, along with logical operators such as \$and and \$or.

Overall, this project helped me understand how MongoDB can be used to manage and query basketball match data efficiently. The project also provided practical understanding of CRUD operations, filtering, sorting, limiting, searching, and counting documents in a NoSQL database.